Sampling can destabilize a continuous-time design

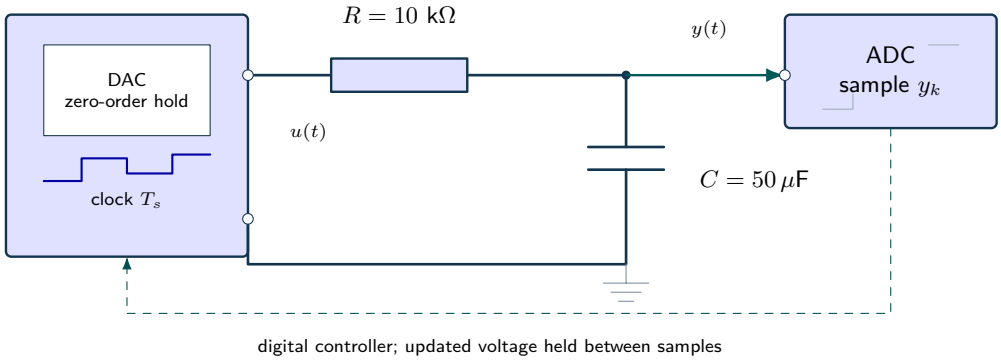
Original learning collection – November 2026

Evidence note. Original teaching scenario. Numerical parameters are deliberately chosen for learning. All generated data and curves are synthetic; no physical measurement is asserted.

1 Design brief

For $G(s) = 1/(0.5s + 1)$, use PI gains $k_p = 2$ and $k_i = 4$ with a held actuator. Apply $u_k = 2(r_k - x_k) + I_k$ before updating $I_{k+1} = I_k + 4T_s(r_k - x_k)$. Derive the exact sampled plant update and two-state closed-loop matrix. Compare $T_s = 0.05$, 0.5 and 1.0 s and find the first discrete-time stability boundary.

2 Physical system



Continuous energy store

$RC = 0.5$ s; $\dot{y} = 2(u - y)$.
The held voltage drives a real-valued continuous state between samples.

Clocked interface

$y_k = y(kT_s)$.
$u(t) = u_k$ for $kT_s \le t < (k+1)T_s$.
Clock and update order matter.

Ideal converter boundary

No quantization, electrical loading or converter dynamics. Use exact held-input discretization when comparing sample periods.

A held voltage drives an ideal RC network. The capacitor voltage is sampled by an ideal ADC; the software controller updates the DAC once per period. Quantization, converter lag and electrical loading are omitted.

3 Continuous design

The continuous PI characteristic polynomial is $0.5s^2 + 3s + 4 = 0.5(s + 2)(s + 4)$, so its poles are stable. This does not establish the stability of a particular digital implementation.

4 Exact held-input plant

Holding $u[k]$ over one sample interval gives $x[k + 1] = ax[k] + bu[k]$, where $a = e^{-2T_s}$ and $b = 1 - a$. The explicit integral update uses the old sample error, so for zero reference,

$$\begin{bmatrix} x[k+1] \\ I[k+1] \end{bmatrix} = M(T_s) \begin{bmatrix} x[k] \\ I[k] \end{bmatrix}, \quad M = \begin{bmatrix} a - 2b & b \\ -4T_s & 1 \end{bmatrix}.$$

Changing the order of integral update and actuator computation would change this matrix.

5 Unit-circle test

Discrete stability requires every eigenvalue of M to lie strictly inside the unit circle. With trace $t = 3a - 1$ and determinant $d = 3a - 2 + 4T_s(1 - a)$, the second-order Jury conditions are $1 - t + d > 0$, $1 + t + d > 0$,

and $1 - d > 0$. For $T_s > 0$, the first expression is $4T_s(1 - a) > 0$. The determinant becomes one at $T_s = 0.75$ s, where the conjugate pair lies on the unit circle. The remaining conditions hold up to that crossing. Thus the first stable interval is $0 < T_s < 0.75$ s, and the endpoint is excluded.

6 Evidence and limits

Plot samples explicitly rather than connecting them as if they were continuous measurements. These results assume exact model parameters, an instantaneous computation at each sample, and no extra communication delay or actuator saturation.

7 What to learn from the experiment

A stable continuous controller can fail after discretization. State the sampling rule and update order, then test the actual discrete closed-loop model.

8 System sketch

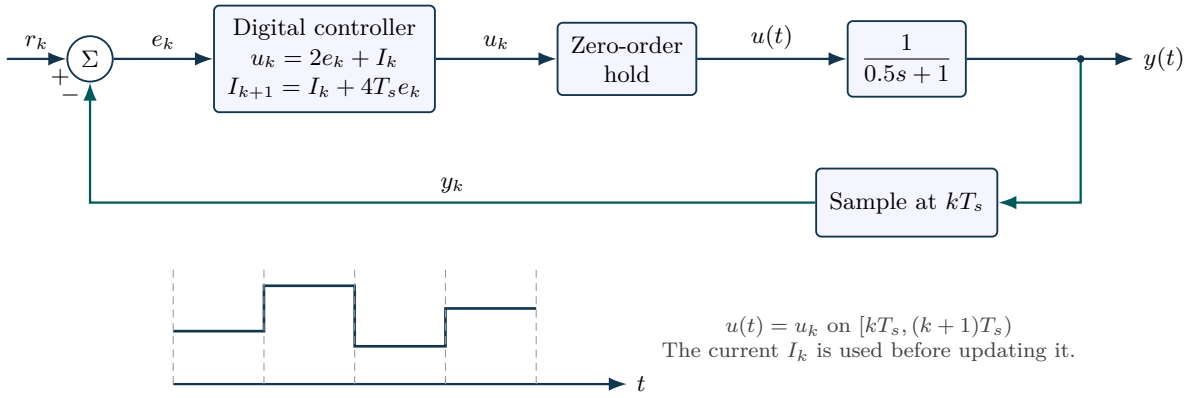
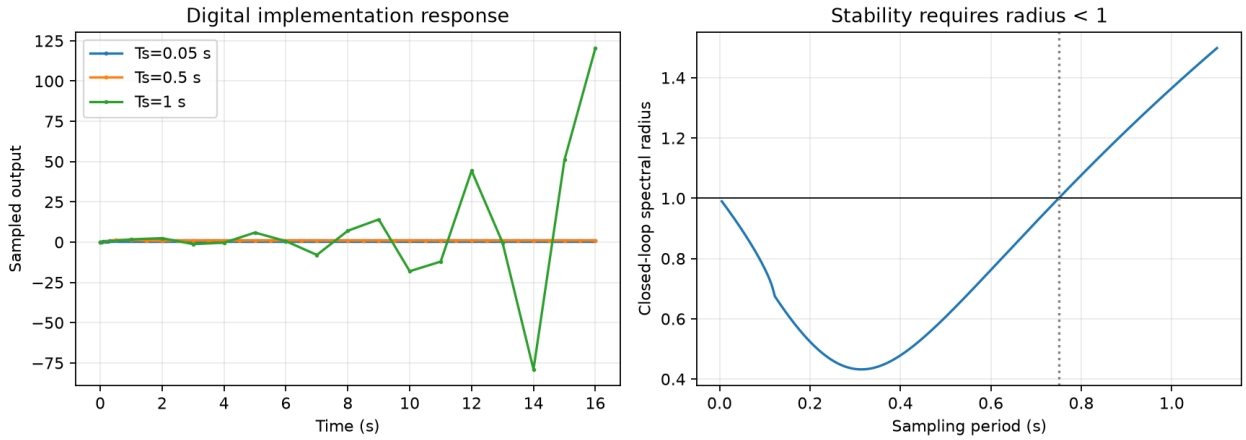


Figure 1. The sampler, digital update order and hold are explicit. Continuous plant dynamics and sampled controller states occupy different parts of the loop.

9 Executed computational evidence

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Ts=0.05 s: spectral radius=0.89390714
Ts=0.50 s: spectral radius=0.60653066
Ts=1.00 s: spectral radius=1.36552727
First unit-circle crossing: 0.75000000 s
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Method references. The equations use standard balance laws and feedback-control methods. Further explanation: https://apmonitor.com/pdc/ and https://fbsbook.org/. Implementation documentation: https://python-control.readthedocs.io/en/0.10.2/. See the repository reference catalog for individual topics and videos.